CROSSSAFE · REGULATORY REGISTER

CrossSafe regulatory and standards register

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DOCUMENT CONTROL

Document ID	CRS-REG-001	Version	0.1
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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/crosssafe		
Source	docs/07-regulatory.md @ d9f28be		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	First register (kit 1.9.1 upgrade); nothing verified by a qualified reviewer; unknowns marked to assess

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Which standards and regulations apply, where, and what is needed. The design follows the US FHWA RRFB interim approval (IA-21) for its form; every other row is "to assess" until a qualified reviewer and the pilot road authority have been asked. Not legal advice; CrossSafe is a research and educational prototype and makes no claim of compliance with any standard. Method: STANDARDS section 23. Sources and the date they were checked are listed at the end.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	FHWA Interim Approval IA-21, rectangular rapid flashing beacons	United States	Indication size (at least 5 x 2 in), SAE J595 class 1 yellow intensity, wig-wag pattern at 75 sequences a minute, mounting	Design basis of PR-001; whether IA-21 is still in force after the MUTCD 11th Edition is not checked	Design lead	To assess
G2	Manual on Uniform Traffic Control Devices (MUTCD)	United States	Sign designs, beacon rules, mounting heights, state supplements	Local rules decide colours, pattern and heights	Pilot road authority	To assess
G3	ADA Standards, operable parts (section 309)	United States	Operable force of 5 lbf (22.2 N) or less, reach range	Push button 0.9 to 1.1 m, 22 N or less (PR-011)	Design lead	Met by design (button not built)
G4	Public right-of-way accessibility guidelines (accessible pedestrian signals)	United States	Audible and tactile pedestrian signals	Button tone and tactile arrow (PR-011)	Design lead	To assess
G5	ASTM D4956, retroreflective sheeting for traffic control	United States	Sheeting classes and performance	The sign sheeting is bought to the local authority's design (CRS-DEC-001)	Pilot road authority	To assess
G6	AASHTO structural supports specification for highway signs, luminaires and traffic signals	United States	Wind loads and support design	40 m/s gust screening in CRS-CAL-001 section G	Structural reviewer	To assess
G7	47 CFR Part 15, unlicensed radio	United States	902 to 928 MHz radios	The sizing assumes 868 MHz; a US pilot needs a 915 MHz radio and a recheck of the link budget	Design lead	To assess
G8	ETSI EN 300 220, short range devices	European Union	Sub-GHz radio, duty cycle (1 % in the 868 MHz sub-band)	Duty cycle 4.3 s against 36 s per hour (STEM check S8)	Design lead	Met by design (paper)
G9	EN 12368 (traffic signals) and EN 12899-1 (fixed vertical signs)	European Union	Signal heads and signs	Applicable only to an EU pilot	Pilot road authority	To assess
G10	IEC 60529, IP code	International	Ingress protection, IP65	Enclosure and connectors IP65 or better (PR-008)	Design lead	Met by design (datasheet)

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G11	UN 38.3 and IEC 62619, lithium batteries	International	Transport tests and safety of industrial lithium cells	The pack is bought; its certificates are to be requested	Design lead	Certification needed later
G12	RoHS Directive 2011/65/EU	European Union	Hazardous substances in electronics	Bought electronics; own board not designed	Design lead	To assess
G13	IEC 62471, photobiological safety of lamps	International	Blue light and retinal hazard classes	LED heads at 3.9 kcd peak; flash area is hazard H5	Design lead	To assess
G14	Privacy and data protection law of the pilot jurisdiction	Pilot jurisdiction	Personal data	No images or audio; counts only (PR-013)	Design lead	Does not apply by design; confirm with the authority
G15	Local road authority rules and permits	Pilot jurisdiction	Colours, flash pattern, heights, installation permit, pole permission	Decides whether the device may go on a public road	Pilot road authority	To assess

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- FHWA, Interim Approval IA-21, 20 March 2018, as cited in CRS-PRB-001; current status not checked.
- U.S. Access Board, Chapter 3: Operable Parts, checked 2026-10-31.
- All other rows name a standard from general knowledge; no link was checked and the edition and clause numbers are not given. They are "to assess" for that reason.